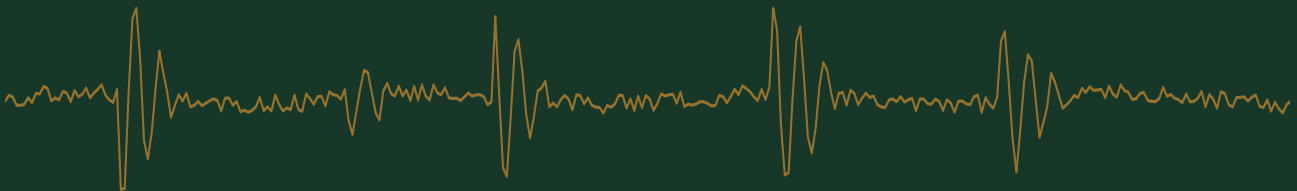




OPEN CATEGORY · THEME 1 — AGRICULTURE

We hear the threat the tree *can't show*.

An autonomous, solar-powered robotic guardian that detects Red Palm Weevil inside the trunk and treats only the trees that are genuinely sick — no internet, no operator.



RPW FEEDING SIGNAL · 2.4 KHZ

Acoustic signature of Red Palm Weevil larvae inside the trunk

Team Vcoders

UNIVERSITY OF PETRA · COLLEGE OF IT · AMMAN, JORDAN

WORLD ROBOT CASPIAN CUP · BAKU

27 JUNE 2026 · PALM-WEB.VERCEL.APP

What's inside.

01	Team Introduction & Roles	03
	Who built Palm Guard, and how the work was divided	
02	Project Idea Summary	04
	The pest, the gap, and what Palm Guard does about it	
03	The Robotic Solution	05
	Origin · research · the autonomous loop · mechanical structure · architecture · sensing · on-device intelligence · decision logic · autonomous treatment · connectivity · challenges	
04	Social Impact & Innovation	15
	The need · a real-world use example · responsible design · key innovation · business model · product ecosystem · roadmap	
—	References	18

THE CORE IDEA

Palm Guard is not just a detector. It closes the robotic loop — **sense → decide → act → alert** — hearing the threat the tree can't show and treating only the trees that are genuinely sick, with no operator and no internet.

4 SENSORS FUSED ON-DEVICE	0–100 RISK SCORE, FULLY OFFLINE	~26 JOD PER PROTOTYPE UNIT	4 AWARDS & FINALIST PLACEMENTS
---	---	--	--

Three engineers, one autonomous guardian.

Team **Vcoders** is three undergraduate engineers from the College of Information Technology at the University of Petra in Amman, Jordan, competing in the Open Category under Theme 1 — Technological Solutions for Agriculture. Between us we cover embedded hardware, applied machine learning, full-stack software and commercialisation, which is what let us take Palm Guard from a research idea to a field-mounted prototype that already runs on solar and a Bluetooth mesh, with the full detect-and-treat loop validated on the bench and on real recordings of the weevil.

We split the work by strength but reviewed every major decision together. Our coach, **Dr. Abedal-Kareem Al-Banna** (University of Petra), guided us on process, planning and logistics only; the construction, programming, this report and the booth were built entirely by the team.

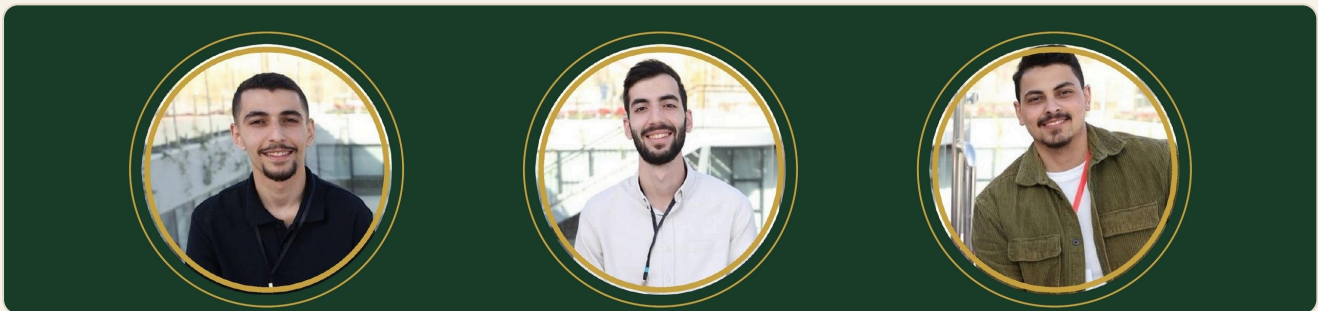
RECOGNITION

Best Innovation — 7th Int'l Jordanian Dates Festival

1st place — IEEE Sustainable Technology Hackathon

2nd place — AAU 2026 Robotics & AI

Finalist — IEEE Region 8 Humanitarian



Abdelrahman Alkurdi

CTO · TECHNICAL LEAD

Hardware, firmware, on-device AI, enclosure CAD and the injection mechanism.

Abdelrahman Al-Haimouni

COO · OPERATIONS LEAD

System integration, field testing, Bluetooth-mesh networking, dashboard and backend.

Zaid Abu Al-Shaar

CBO · BUSINESS LEAD

Problem research, social-impact analysis, business model and presentation.

Figure 1. Team Vcoders — University of Petra, College of Information Technology.

By the time you can see it, the tree is lost.

The pest, the gap, and what Palm Guard does about it.

The Red Palm Weevil is the most destructive palm pest in the world, and it kills from the inside: the female lays her eggs in the trunk, the larvae tunnel outward unseen, and by the time any external symptom shows the tree is usually beyond saving. It has invaded forty-nine countries and is classified by the FAO as the most dangerous palm pest of all. The damage is done before anyone can see it — exactly the gap Palm Guard closes.

Palm Guard is a low-cost, solar-powered, internet-free robotic device that clamps onto the trunk and runs a complete **sense, decide and act** loop on its own. A steel probe carries four sensors into and against the trunk; the device listens continuously for larval feeding sounds and senses boring vibration, internal temperature rise and the gas signature of decaying tissue. An on-device AI ensemble fuses these signals into a single **risk score from 0 to 100**. When an infestation is confirmed, the device does not merely raise an alarm: it activates a peristaltic pump and cannula to deliver a controlled trunk-injection treatment pathway, then reports the event over a Bluetooth mesh to a live dashboard. For the WRCC booth demonstration, the same pump pathway is shown using **clear water only**.

The value is early, targeted, autonomous intervention in place of late visual detection and blanket spraying: Palm Guard catches an infestation weeks before symptoms appear, treats only the trees that need it, and replaces repeated manual scans with one permanent device.

ALREADY BUILT & BENCH-VALIDATED

Firmware on the ESP32-S3 with a model trained on real weevil audio; a prototype mounted on a palm and running on solar and a Bluetooth mesh; and a complete v4 enclosure and injection system at roughly **26 JOD per unit**. Validating the full loop against a live field infestation is the next step.

Existing tools stop at “there may be a weevil.” Palm Guard detects, decides *and* treats — a permanent guardian that hears what the tree cannot show.

A clamp-on robot that senses, decides and acts.

A low-cost, solar-powered, internet-free device that runs the whole loop on its own.

Palm Guard clamps onto the trunk and carries four sensors on a steel probe into and against the wood. It listens continuously for larval feeding, and senses boring vibration, internal temperature rise and the gas signature of decaying tissue. An on-device AI ensemble fuses these into a single **risk score from 0 to 100**. When an infestation is confirmed it does not merely raise an alarm — it drives a peristaltic pump and cannula to deliver a controlled trunk-injection treatment, then reports over a Bluetooth mesh to a live dashboard. For the WRCC booth, the same pump pathway runs **clear water only**. The value is early, targeted, autonomous intervention in place of late visual detection and blanket spraying — catching an infestation weeks before symptoms appear and treating only the trees that need it.



Figure 2. The v4 device — a clamp-on node with a forward-tilted solar panel, an RGB status light, and the steel probe and treatment cannula penetrating the trunk. The prototype is built and bench-validated, running on solar and a Bluetooth mesh; validating the detect-and-treat loop against a live infestation is the next field step.

Built & bench-validated FIRMWARE ON ESP32-S3 · MODEL ON REAL WEEVIL AUDIO	~26 JOD / unit COMPLETE V4 ENCLOSURE + INJECTION SYSTEM	Solar + BLE mesh OFF-GRID, INTERNET-FREE OPERATION
--	--	--

The autonomous loop — decide, then act.

Not a fixed-sequence machine: it acts only when the evidence converges.

01 ● Sense Four sensors read the trunk continuously.	02 ● Extract MFCC + spectral + environmental features.	03 ● Score Ensemble → risk 0–100.	04 ● Agree Threshold + multi-sensor agreement.	05 ● Act Inject & report — or keep monitoring.
--	--	---	--	--

We started from one hard fact: with this pest, by the time you can see the problem the tree is usually lost, so early detection from inside the trunk is the whole game. Two decisions shaped Palm Guard. First, rather than trusting one signal, we fuse **four independent inside-trunk signals** with on-device AI to suppress false alarms. Second — and this is what makes Palm Guard a robot rather than a sensor — we **close the loop**: the device decides, then acts on its own by injecting treatment, with no operator and no internet.

HOW PALM GUARD DIFFERS

Every competitor stops at detection.

CRITERION	PALM GUARD	PALMEAR	IOTREE (AGRINT)	TRAPS / VISUAL
Detection method	Acoustic + vibration + thermal AI	Acoustic + AI	Seismic sensor	Visual / traps
Pre-symptom detection	✓	✓	✓	✗
Continuous 24/7	✓	✗	✓	✗
Autonomous, no manual scan	✓	✗	✓	✗
Automatic treatment of the tree	✓	✗	✗	✗
Solar / off-grid	✓	✗	~	✗
Connectivity	Bluetooth mesh	App per scan	Cloud gateway	—

One permanent device replaces repeated scans and labour for a lower total cost of protection. To our knowledge, no existing product offers the autonomous treatment stage.

Two zones and a head — a clean drop-in build.

A self-developed edge device — not an off-the-shelf kit.

Zone A faces the palm and carries a steel probe inserted into the trunk, an acoustic sound guide, the microphone, the temperature, vibration and gas sensors — the gas sensor in its own sealed chamber so it reads only the trunk's internal atmosphere.

Zone B holds the electronics: the ESP32-S3 controller, the power chain (LiPo, CN3065 solar charger, TPS63802 regulator), the peristaltic pump, reservoir and cannula, and the RGB status light. The **head** carries a forward-tilted solar panel aimed at the sun.

The v4 design is mechanically complete: closed base, real sensor openings, corrected solar orientation, verified for interference and sealing, split into three printable parts (Body, Lid, Probe) with full STLs and a build manual.

Prototype BOM ≈ **26 JOD**.

DESIGN PRINCIPLE

Engineering care, not just control — an acoustic waveguide couples the inside of the trunk to the mic so faint feeding sounds reach the sensor cleanly; the probe and the treatment cannula both penetrate the trunk.

TABLE 1 · PROTOTYPE BILL OF MATERIALS BY SUBSYSTEM — ≈ 26 JOD AT PROTOTYPE QUANTITIES; TARGET ≈ 10 USD / UNIT AT SCALE

CONTROLLER	ESP32-S3 — WiFi and Bluetooth LE on-chip
SENSING	INMP441 acoustic mic · SW-420 vibration · DS18B20 waterproof temperature · BME680 gas
POWER	LiPo 3.7 V · 5 V solar panel (135×64.7 mm) · CN3065 solar charger · TPS63802 buck-boost (3.3 V) · 220 µF capacitor · KCD1-105 switch
ACTUATION	5 V micro-peristaltic pump · IRF540 MOSFET (33 A / 100 V) · 2×4 mm silicone tube · RGB status LED
COMMS	Bluetooth LE mesh — built into the ESP32-S3, no separate radio module
BUILD	Double-sided PCB, 9×15 cm

Inside the device — and the stack behind it.

A clean drop-in enclosure, and a self-developed software / hardware stack.

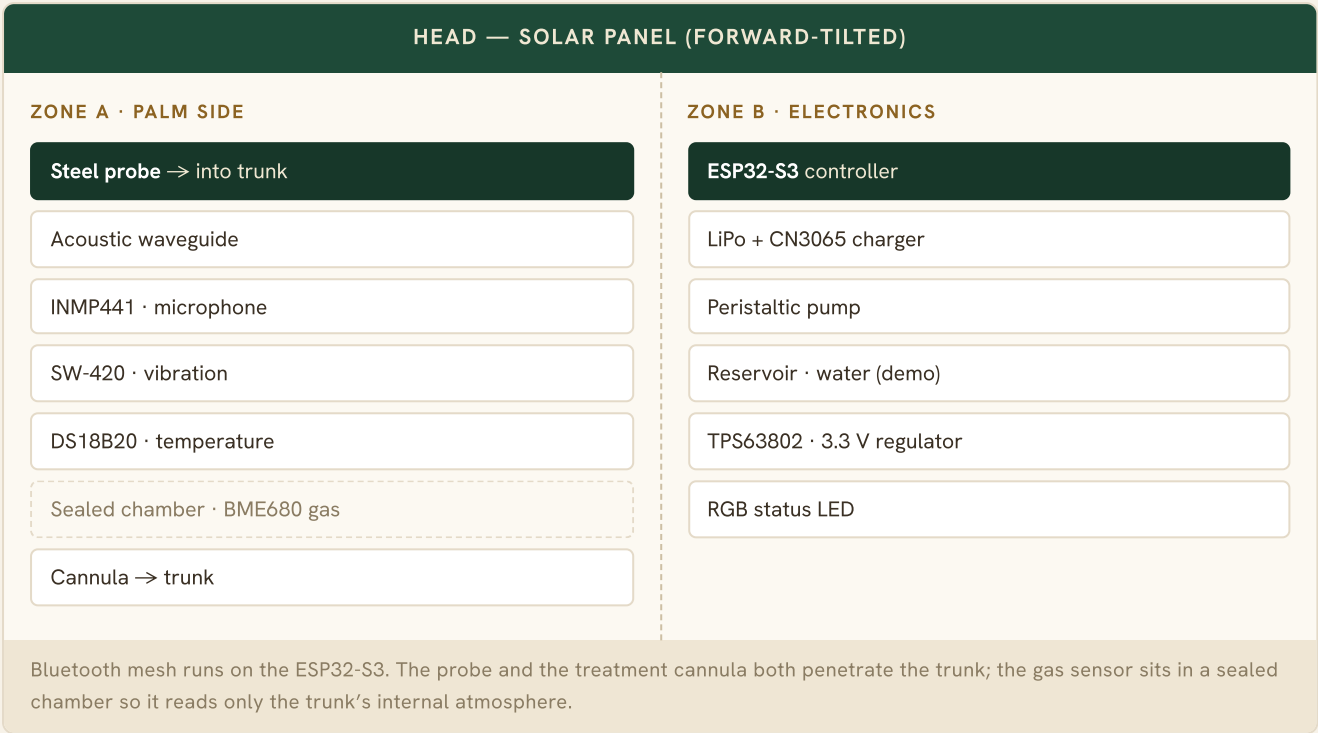


Figure 3. Enclosure section — Zone A couples to the trunk through a steel probe and an acoustic waveguide and carries the four sensors; Zone B holds the controller, power, pump, reservoir, regulation and the status LED.

SYSTEM ARCHITECTURE

A custom multi-model pipeline, not a kit.

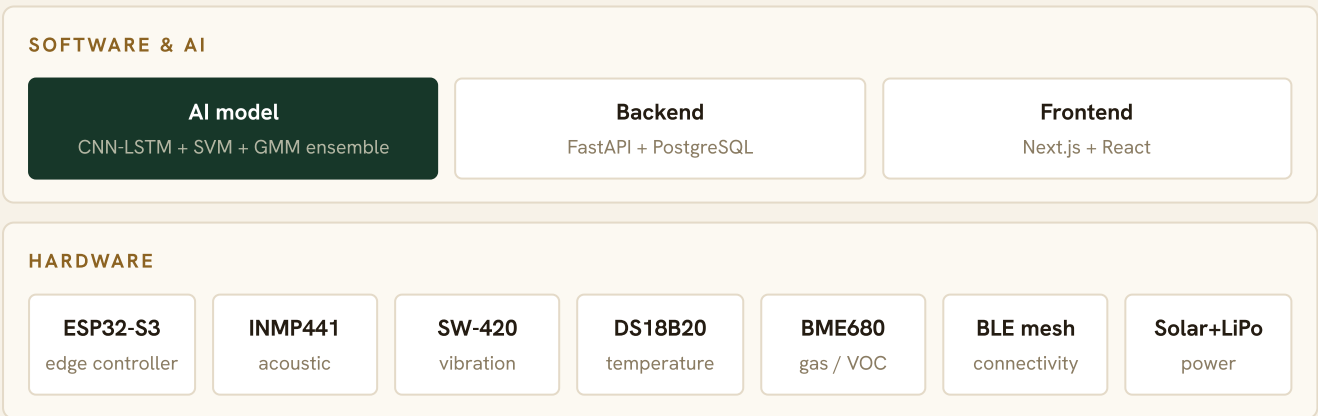


Figure 4. System architecture — a self-developed edge device with a custom multi-model pipeline, not an off-the-shelf kit.

Four senses inside the trunk, fused on-device.

INMP441 · MICROPHONE

Low-amplitude larval feeding and boring sounds inside the trunk — energy in the low band with peaks near 2.4 kHz.

SW-420 · VIBRATION

The mechanical vibration produced as larvae bore through the wood.

DS18B20 · TEMPERATURE

Internal trunk-temperature rise from larval metabolic activity.

BME680 · GAS

The gas and odour signature of frass and fermenting tissue in an infested trunk.

WHY FOUR, NOT ONE

A single microphone confuses wind with feeding. Requiring agreement across four independent signals lets the model confirm an infestation only when the evidence converges — which sharply reduces false positives, the main weakness of single-microphone detectors.

WIRING OVERVIEW

Everything on one controller.

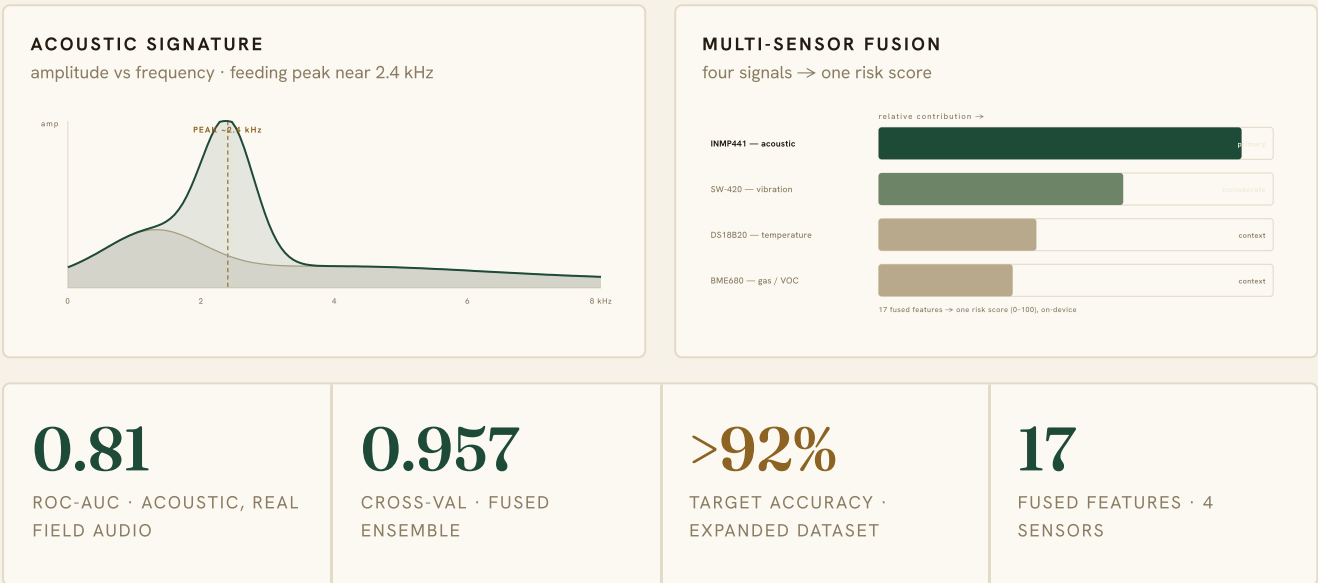


Figure 5. Wiring overview — every sensor and actuator connects to the ESP32-S3 over a standard bus; the treatment pump is driven from GPIO38 through an IRF540 MOSFET.

A hybrid ensemble, running at the edge.

Inference on the ESP32-S3 — offline, low-latency, low-power and private.

Audio is framed and converted to MFCC and spectral features; dynamic thresholding and noise reduction clean the signal; and a hybrid ensemble combines three complementary models — a **Gaussian Mixture Model** for acoustic background and anomaly, a **Support Vector Machine** for feature classification, and a **CNN-LSTM** for the spectro-temporal pattern of feeding bursts. The acoustic result is fused with the vibration, temperature and gas features — seventeen in total — into one risk score.



Measured on-device with full offline inference. These are early prototype validation results — they confirm the sensing and classification pipeline works; larger multi-farm testing is needed before claiming commercial-level accuracy.

The loop, as compiled firmware.

Listing 1 — PlatformIO / C++ running locally on the ESP32-S3, published open-source by the team.

LISTING 1 · SENSE → DECIDE → ACT LOOP — ESP32-S3, FIRMWARE V3.0

```
// Palm Guard — sense → decide → act loop (ESP32-S3, firmware v3.0)
void loop() {
  SensorFrame f = readSensors();           // INMP441, SW-420, DS18B20, BME680
  float feat[17];
  extractFeatures(f, feat);                // MFCC + spectral + environmental
  float risk = ensemble.score(feat);       // GMM + SVM + CNN-LSTM → 0..100
  if (risk >= RISK_THRESHOLD && sensorsAgree(f)) {
    injectTreatment();                     // GPIO38 → IRF540 → pump; demo = clear water
    mesh.send(buildAlert(risk, f));         // report over Bluetooth mesh
    setStatus(ALERT);                       // status LED → red
  } else {
    setStatus(MONITORING);                  // status LED → green
  }
  sleepUntilNextWindow();                  // duty-cycle for solar / battery
}
```

EMBEDDED ON THE ESP32-S3

The full pipeline — MFCC features, the GMM/SVM/CNN-LSTM ensemble and the risk score — is compiled to run on the microcontroller itself, so inference happens at the edge with no cloud call.

ACTS WITHOUT A NETWORK

When the score crosses threshold and multiple sensors corroborate, the device injects and reports over Bluetooth mesh — with no human and no internet in the path.

PUBLISHED

The full firmware and the live dashboard are published by the team. Live monitoring: palm-web.vercel.app.

It treats only when the evidence agrees.

The conditional decision-making that makes this a robot, not a repeating machine.

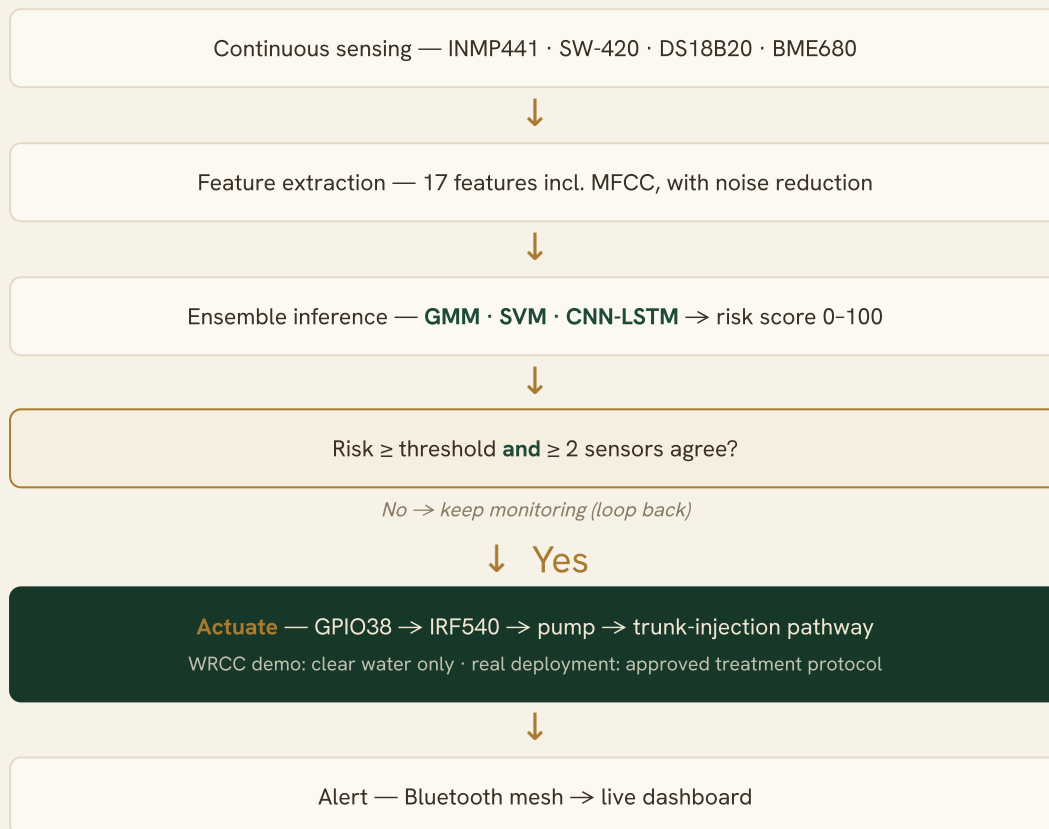


Figure 6. On-device decision logic — the device treats only when the risk score crosses threshold and multiple sensors corroborate, then acts and reports without any human or network.

WHY THIS IS A ROBOT, NOT A SENSOR

Every competitor stops at “there may be a weevil.” The conditional branch above — act *only* on agreement, otherwise return to monitoring — is what makes Palm Guard an autonomous robotic node: it senses, decides, acts and reports on its own.

Closing the loop — the part that sets us apart.

When an infestation is confirmed, the ESP32-S3 drives a control line through an IRF540 MOSFET to a 5 V peristaltic pump, moving liquid from the onboard reservoir through a cannula into the trunk beside the sensing probe. The device senses, decides, acts and reports without remote control, internet or a human operator. Trunk micro-injection of emamectin benzoate is an **established** preventative and curative method — a field trial in Jordan by the National Agricultural Research Centre confirmed its effect. Palm Guard's contribution is delivering a proven treatment pathway **automatically and early**, the moment the on-device model confirms risk, and logging every event for traceability.

FUNCTION	WRCC BOOTH DEMONSTRATION	REAL AGRICULTURAL DEPLOYMENT
Liquid used	Clear water only	Approved trunk-injection treatment
Purpose	Safely show autonomous actuation	Treat a confirmed infestation
Trigger	AI risk threshold / demo pattern	AI risk threshold from field sensors
Human role	No remote control during actuation	Farmer / agronomist supervises setup
Safety	No pesticide, smoke or unsafe liquid	Follows agricultural regulation
Output	Pump · LED · mesh alert · dashboard	Treatment log · alert · history

THE LIVE BOOTH DEMONSTRATION

In the 2×2×2 m booth, judges see the complete sense-decide-act-alert loop run safely: the device streams readings, scores risk, updates the dashboard, then autonomously actuates the pump with clear water as the status LED changes and a mesh alert appears. Because treatment is targeted only to confirmed trees as a precise in-trunk dose, Palm Guard reduces the need for broad insecticide spraying.

Offline-first, solar-powered, always watching.

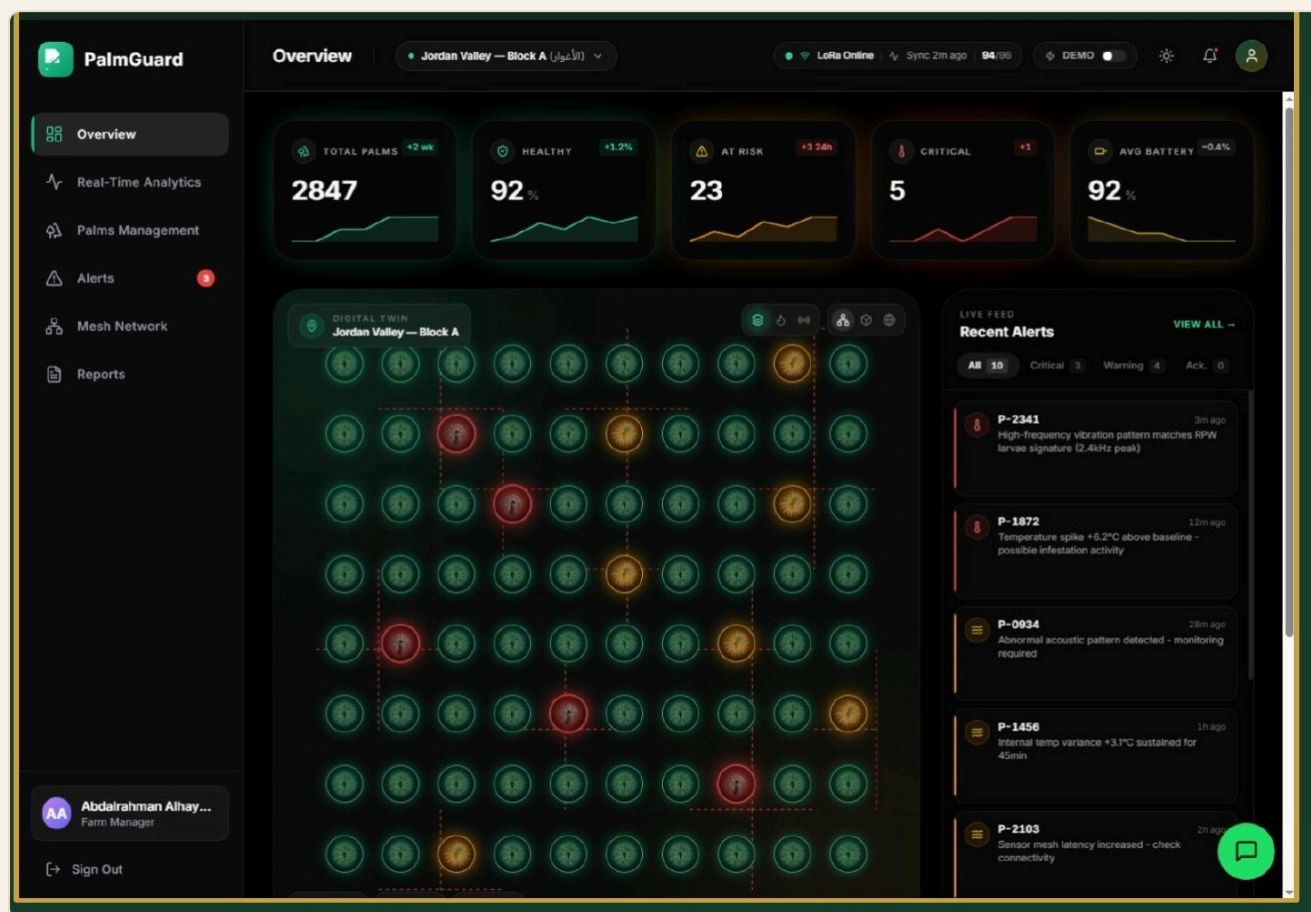


Figure 7. The live monitoring dashboard (palm-web.vercel.app), Jordan Valley deployment view: a per-palm digital twin showing every tree as healthy, at-risk or critical; an AI alert feed surfacing the acoustic, vibration and temperature signatures behind each flag; and a mesh & device-health view tracking battery and solar status across the network.

POWER & CONNECTIVITY

The ESP32-S3's built-in Bluetooth LE links units in a low-power mesh and forwards events to a nearby gateway with no cellular network — the system is offline-first.

A forward-tilted 5 V solar panel, LiPo, CN3065 charger and TPS63802 regulator give effectively zero grid draw and continuous round-the-clock operation.

The dashboard presents a farm digital twin, real-time alerts with the signal behind each flag, and a mesh & device-health view across the network.

CHALLENGES WE SOLVED

- Tiny insect sounds vs ambient noise → multi-sensor fusion, MFCC features and dynamic thresholding
- Running an ensemble on a microcontroller → model compression to fit the ESP32-S3
- Trunk coupling & sealing → resolved in the v4 drop-in assembly
- Off-grid autonomy → solar power and low-power BLE duty cycling
- Safe live demonstration → clear-water booth pathway, separate from real treatment

A pillar of food security, under attack.

49

COUNTRIES INVADED

€483M

DAMAGE / YR ·
MEDITERRANEAN

80%

OF ITS LIFE HIDDEN IN
THE TRUNK

25–40%

HARVEST LOST
UNTREATED

The FAO classifies the Red Palm Weevil as a category-1 pest and the most dangerous palm pest in the world; it attacks more than forty palm species. Because the larvae feed inside the trunk, an infestation stays invisible until the tree is dying — and a mature date palm is six or more years of growth, worth on the order of a thousand US dollars. For Jordan the threat is immediate: the weevil reached the Jordan Valley around 1999, and at the Azraq oasis (~50,000 palms) about 1,000 infested trees were removed in a single campaign. Critically, the trunk-injection treatment Palm Guard automates was validated in Jordan itself.

Instead of spraying an entire grove, Palm Guard treats only the confirmed, infected tree with a precise micro-dose — *green high-tech* that works with the living system rather than against it.

RESPONSIBLE DESIGN

False positives: needs high risk + ≥ 2 sensor groups; optional human-confirm mode.

Chemical: sealed reservoir, local regulation, agronomy supervision.

Data: offline-first; farm data stays local.

Food

A DIRECT PILLAR OF ARAB
FOOD SECURITY

Economic

DECADES OF PRODUCTION
LOST PER TREE

Social

PRIMARY INCOME FOR
JORDAN VALLEY FARMERS

Environment

PALMS RESIST
DESERTIFICATION

THE KEY INNOVATION

Palm Guard is a low-cost, fully autonomous, internet-free robotic system that both *detects* Red Palm Weevil activity through multi-sensor sensing and an on-device AI ensemble, and *responds* through an automatic trunk-injection pathway. Existing solutions mainly stop at detection; Palm Guard closes the loop.

From signal to a saved tree.

What the loop looks like on a real farm — and the platform behind it.

Consider a date farm in the Jordan Valley. Palm Guard units are clamped to the trunks and form a Bluetooth mesh back to a single gateway, and the farmer watches a live dashboard of every tree.

On one palm, the device picks up a high-frequency vibration and acoustic signature consistent with larval feeding — a peak near 2.4 kHz — together with a temperature rise above baseline. The fused risk score crosses the confirmation threshold, the device triggers a controlled trunk-injection treatment according to the approved field protocol, and an alert appears on the dashboard. The tree is treated **weeks before** any external symptom would have shown — where today's practice would catch it only when the fronds begin to collapse, too late to save.

04 · PRODUCT ECOSYSTEM

A connected platform, not a single device.

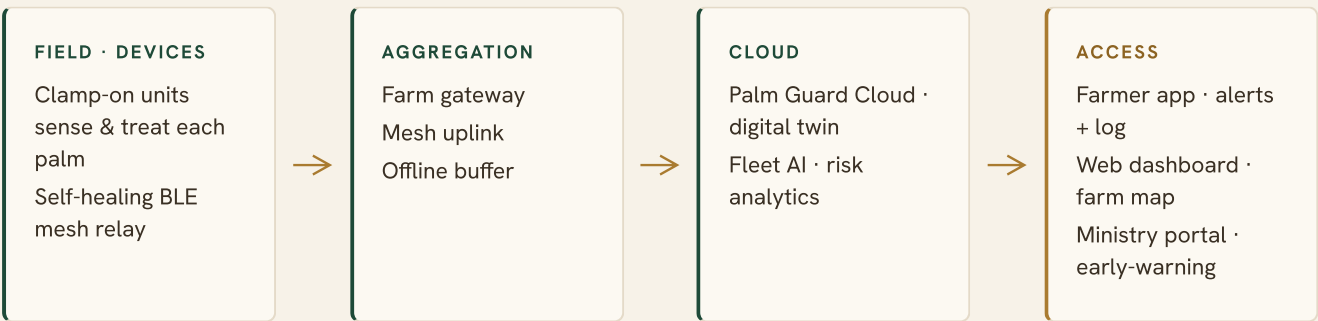
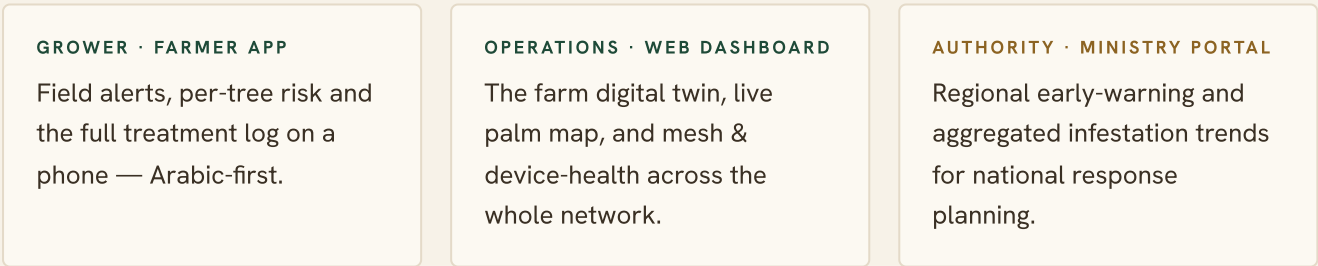


Figure 8. The Palm Guard platform — data flows from autonomous clamp-on units up through a self-healing Bluetooth mesh and farm gateway to the cloud digital twin, then out to the farmer app, the web dashboard, and a ministry early-warning portal.



WHY A PLATFORM, NOT A PRODUCT

One device protects a tree; the mesh protects a region — per-tree detection becomes farm-wide and ultimately national early warning, with farm data kept local and offline-first.

Built to become a real product.

~10 USD to make · 20 USD hardware + subscription · 200–350 USD saved per tree.

1 CUSTOMER SEGMENTS

- Small-large palm farms
- Municipalities & Ministry of Agriculture
- Landscaping & resorts
- Gulf high-value date farms

2 VALUE PROPOSITION

- Detection weeks before symptoms
- **\$200–350** saved per tree
- Prevents **70%+** tree loss
- First local detect-and-treat in Jordan

3 CHANNELS

- Direct field sales
- Government pilot → national
- Exhibitions & unions
- WhatsApp + on-farm demos

4 RELATIONSHIPS

- Automated alerts & monitoring
- Farmer support line
- Annual maintenance visits
- Training workshops

5 REVENUE STREAMS

- Hardware **\$20** / device
- Subscription \$30 / \$75 / \$150 mo
- Install & maintenance
- Data insights · Gulf tier

6 KEY RESOURCES

- Acoustic / vibration / thermal AI
- BLE-mesh firmware
- ESP32-S3 device
- Cloud dashboard & risk engine

7 KEY ACTIVITIES

- Edge data collection
- Model training & updates
- Install & mesh deployment
- Field support

8 KEY PARTNERS

- Ministry of Agriculture
- Farms & cooperatives
- Hardware suppliers
- Universities & research

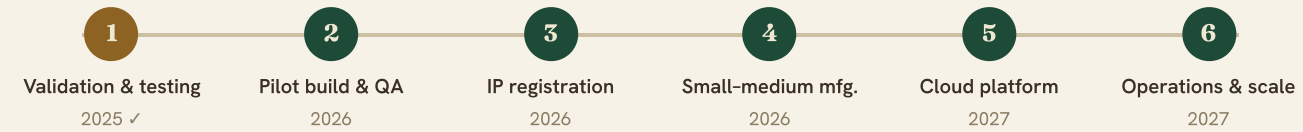
9 COST STRUCTURE

- Manufacturing **~\$10** / unit
- Cloud & AI training
- Field ops & install
- Maintenance & R&D

UNIT ECONOMICS

Today's prototype BOM is ~26 JOD because it is built in small quantities from off-the-shelf parts; at scale the target manufacturing cost is ~10 USD per unit through bulk purchasing, PCB integration and simplified assembly — against value to the farmer of 200–350 USD per tree saved and prevention of more than 70% of tree loss.

From validated prototype to operational scale.



Stage one is complete — firmware, on-device AI, dashboard and the v4 enclosure and injection system are built and award-validated. Near-term priorities are to expand the real-world weevil audio dataset past **92% accuracy**, run a Jordan Valley field pilot, ruggedise the enclosure, validate treatment dosing and obtain regulatory approval, and scale the mesh while driving unit cost down. The path runs in three stages: a competition prototype that safely demonstrates the full loop with clear water; a field pilot to expand the dataset and validate the live-infestation response; and regulated deployment with approved treatment, maintenance teams and ministry or farm partnerships.

FUNDING & NEXT MILESTONE

The team is raising a **15,000 JOD pre-seed** for pilot manufacturing, IP registration and testing, on a path to seed readiness and aligned with the global coalition against the Red Palm Weevil. The next technical milestone is **live-infestation validation**: testing the full detect-decide-act loop on confirmed infested palms under controlled agricultural supervision.

REFERENCES

[1] FAO. **The Red Palm Weevil threatens food security and livelihoods**. FAO Newsroom, 2019.

[2] FAO. **Current Situation of RPW in the NENA Region** — Scientific Consultation, Rome, 2017.

[3] CABI Plant Health Cases. **Management of RPW in Al-Ahsa Oasis**, 2023.

[4] The Jordan Times. **Campaign launched to eliminate palm tree pest**, 2025.

[5] Soroker, V. et al. Phytoparasitica. **Current status of RPW infestation in date palm**.

[6] Mashal, M. M. & Obeidat, B. F. (2019). **Efficacy of emamectin benzoate micro-injection to control RPW**. Heliyon 5(6), e01833. NARC, Jordan.

[7] **Early detection of RPW using a distributed optical sensor**. Scientific Reports, 2020.

[8] **Feeding or starvation: RPW susceptibility to insecticides**. Frontiers in Insect Science, 2025.

[9] Earth Island Journal. **How AI Could Save Date Palms**, 2024.

[10] **Early Detection of RPW using Deep Learning of Acoustic Signals**. arXiv:2308.15829, 2023.

[11] Koubaa, A. et al. (2020). **Smart Palm: An IoT Framework for RPW Early Detection**. Agronomy 10(7).

[12] Wang, B. et al. (2021). **Detecting RPW Using ML and Fiber-Optic Distributed Acoustic Sensing**. Sensors 21(5).

[13] EcoMENA. **Red Palm Weevil: A Real Threat to Date Palms in the Middle East**, 2025.